

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. ECE A

Course Code & Title: EC6703 Embedded and Real Time Systems

Name of the Faculty member: Mr.T.Ramprakash

Name of the Topic: Introduction to Embedded Computing

Name of the Innovative Practice: Brainstorm

Date & Time: 05.07.2019 & 30 Minutes

ICT Tool Used: LCD projector, Personal Computer

Description:

Brainstorming combines a relaxed, informal approach to problem solving with lateral thinking. It encourages people to come up with thoughts and ideas that can, at first, seem a bit crazy. Some of these ideas can be crafted into original, creative solutions to a problem, while others can spark even more ideas. This helps to get people unstuck by "jolting" them out of their normal ways of thinking.

Goals (Learning Outcomes):

Learning Outcomes:

1. The students will be able to explain the fundamental concepts of designing Embedded and Real Time Systems.
2. The students will be able to acquire better knowledge in developing an embedded system.

Use of appropriate methods:

Justification for choosing Brainstorm Activity:

Students will get interested if a real time application is taken and discussed. Students will give their own ideas and they will discuss the same to other students.

Effective presentation

Details of the Implementation:

To make clear about the embedded system application, we took an application of embedded system – “Conference Room Automation”.

Problem Statement:

A conference room should be automated with the following requirements.

1. As soon as the first person enters the conference room, all the appliances like light, projector and AC should be switched on.

2. The system should count the persons as they enter in to the conference room.
3. After a mean time, the system should send the count of the members in the conference room to cafeteria for preparation of snacks and tea.
4. After completion of the meeting, after the last person leaves the hall, all the appliances should be switched off.

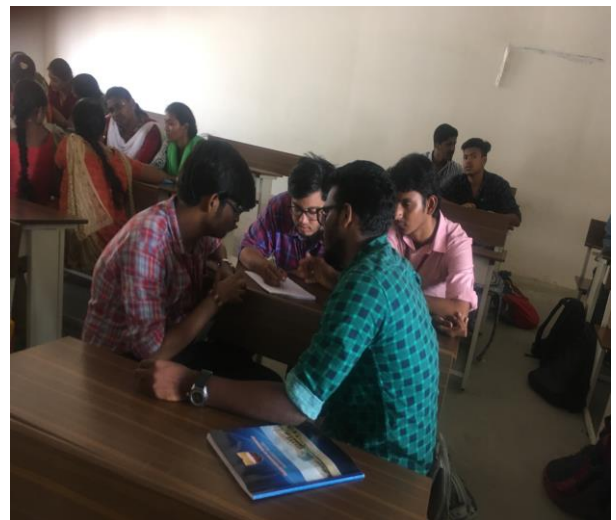
Solutions:

Four Students were clubbed in to groups and were given time to discuss to come out with solutions for the above problem. All the group though in a different manner and came with unique ideas to implement the product.

1. First team suggested to place a IR sensor in the door to sense the arrival of the member. As soon as the IR sensor detects a person, the processor will increment the count and switch on all the appliance. After some time, the count can be sent to cafeteria by means of GSM attached to the processor.
2. Second team suggested to keep a camera and by using image processing, they can detect the motion of human and accordingly they can switch it on and off the appliance.
3. Third team suggested to use PIR sensor, ultrasonic sensor for the detection of the persons.
4. Fourth team suggested to implement using AI for the same.

Innovative Teaching Method Execution

Introduction to Embedded Computing - Brainstorm



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ While the demo was done live by opening the Task Manager in the personal computer and showing the task, process and thread in it, students were very eager to watch they understood the difference in a very better manner.

Reflective Critique:

- **Benefits:**
 - After completion of the Brainstorming session, all the students voluntarily involved themselves and gave very interesting ideas. Students were live in the entire discussion.
- **Challenges**
 - First 2 teams hesitated and felt nervures while sharing their ideas and later on as all the students involved in the topic the other team presented well without any hesitation.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1					2	3		1

CO1: The students will be able to explain the fundamental concepts of designing Embedded and Real Time Systems

References:

- Marilyn Wolf, “Computers as Components - Principles of Embedded Computing System Design”, Third Edition “Morgan Kaufmann Publisher,2012
- <https://teaching.unsw.edu.au/brainstorming>

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Degree, Semester& Branch: VII Semester B.E. ECE A

Course Code & Title: EC6703 Embedded and Real Time Systems

Name of the Faculty member: Mr.T.Ramprakash

Name of the Topic: Assembly, Linking and Loading

Name of the Innovative Practice: Animated Videos

Date & Time: 29.07.2019 & 06 Minutes

ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

1. The students will be able to acquire better knowledge about the operation of assembler, linker and loader.
2. The students will be able to easily differentiate the working of compiler and assembler.

Use of appropriate methods:

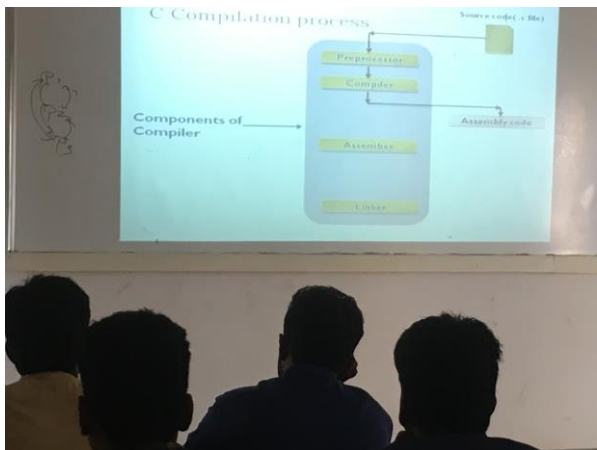
Justification for choosing Videos about Assembly, Linking and Loading:

Many of the students confuse the operation of Assembler, Linker and Loader. They confuse the compiler operation with Assembler and Assembler operation with linker. So if visualization is shown regarding this topic, the confusion would be rectified.

Effective presentation

Details of the Implementation:

The topic was first handled and many of the students were under confusion regarding the working operation of Compiler and Assembler. Then the video was shown and almost all the students were able to relate the working operation of Compiler, Assembler, Linker and Loader.

Innovative Teaching Method Execution	
Assembly, Linking and Loading – Videos	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 60% of students answered correctly.**

Reflective Critique:

- **Benefits:**
 - Some of the students were not able to understand the difference between compiler and assembler. But after watching the video, almost all the students were able to understand the difference of compiler and assembler.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	3	2	1					1	1		

CO 2: The Students will be able to describe the computing required for Real Time Embedded Systems.

References:

- Marilyn Wolf, “Computers as Components - Principles of Embedded Computing System Design”, Third Edition “Morgan Kaufmann Publisher, 2012
- <https://www.youtube.com/watch?v=N2y6csonII4>
- <https://www.geeksforgeeks.org/difference-between-compiler-and-assembler/>

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Degree, Semester& Branch: VII Semester B.E. ECE A

Course Code & Title: EC6703 Embedded and Real Time Systems

Name of the Faculty member: Mr.T.Ramprakash

Name of the Topic: Multitask and Multiple Process

Name of the Innovative Practice: Demonstration

Date & Time: 14.08.2019 & 20 Minutes

ICT Tool Used: LCD projector, Personal Computer

Description:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. A demonstration is the process of teaching someone how to make or do something in a step-by-step process.

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

Goals (Learning Outcomes):

Learning Outcomes:

1. The students will be able to understand the difference between program, task and thread.
2. The students will be able to acquire knowledge on multitasking and multiprocessing.

Use of appropriate methods:

Justification for choosing Demonstration Activity:

Students are not aware of the difference between a program, task, process and thread. Though they knew the definitions of the above, they were not able to identify the difference. So demonstrating a task, process and thread will impart a good knowledge about the above topics.

Effective presentation

Details of the Implementation:

The topic was first handled and the question was raised to the students about a task, process and thread. Most of the students were not able to differentiate it. Then I gave a demonstration using the personal computer and insisted them when to call it as a program, when to call it as a process and what is thread. All the students were able to clearly understand the difference clearly.

Innovative Teaching Method Execution	
Multitask and Multiple Process - Demonstration	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ While the demo was done live by opening the Task Manager in the personal computer and showing the task, process and thread in it, students were very eager to watch they understood the difference in a very better manner.

Reflective Critique:

- **Benefits:**

- After completion of the demonstration, all the students were able to differentiate the definition program, task, process and thread

- **Challenges**

- While demonstrating it for the first time, many of the students found difficult to follow the content, so I again repeated the demonstration to make them clear with new examples. So it took some extra time to finish the demonstration

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	3	3	1					2	1		

CO3: The students will be able to apply the concepts of scheduling in Real Time Operating System.

References:

- Marilyn Wolf, “Computers as Components - Principles of Embedded Computing System Design”, Third Edition “Morgan Kaufmann Publisher,2012
- <https://cft.vanderbilt.edu/2013/03/teaching-demonstrations-advice-and-strategies/>

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Degree, Semester& Branch: VII Semester B.E. ECE A

Course Code & Title: EC6703 Embedded and Real Time Systems

Name of the Faculty member: Mr.T.Ramprakash

Name of the Topic: Design Methodologies

Name of the Innovative Practice: One Minute Paper

Date & Time: 05.09.2019 & 1 Minute

ICT Tool Used: NIL

Description:

A “one-minute paper” may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question, which prompts students to reflect on the day’s lesson and provides the instructor with useful feedback.

Goals (Learning Outcomes):

1. Students will be able to reflect on their understanding about the different types of Design methodologies.
2. The students will be able to easily differentiate the different types of design methodologies and will be able to list the pros and cons of different design methodologies.

Use of appropriate methods:

Justification for choosing using One minute paper activity:

Design methodologies have five different methods. So if an immediate recall is done about the topic in the same class, students will be able to easily follow and will not forget about the five methods in Design Methodologies.

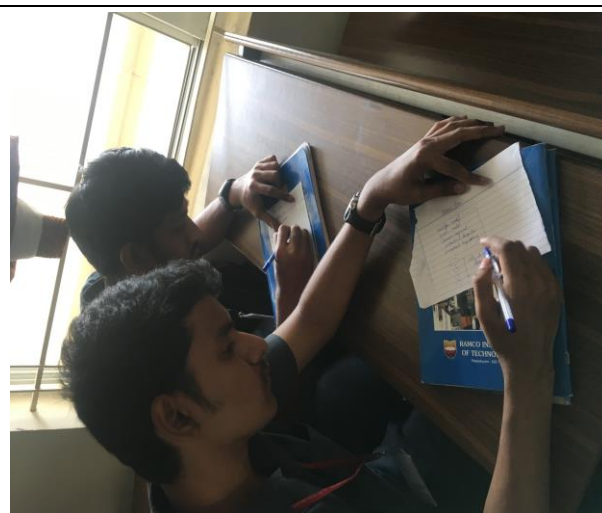
Effective presentation

Details of the Implementation:

The topic was first handled and then the students were asked to summarize the design methodologies that were discussed in the class. Students wrote their understanding of the topic in one minute and handed the paper to me. After reading all the paper, I came the understand how many of the students understood the concept correctly and how many of them had misunderstood. This activity helped me to understand the students understanding on the topic and on the very next day I again summarised the topic and pointed out some places where the students have misunderstood the topic.

Innovative Teaching Method Execution

Measurement of Impedance, Frequency, Power - Demonstration



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that one point was missed with respect to method of computation and finally I insisted the point they missed to tell.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 60% of students answered correctly.**

Reflective Critique:

- **Benefits:**

- Minute papers are an effective way of involving all students in class simultaneously. It gave me the information about how many students understood the concept correctly and how many misunderstood the concept. It ensures equal participation of each and every class member. It can promote class attendance and attentiveness.

- **Challenges**

- I have planned One minute paper activity for 2 minutes but it took 5 minutes because of narrating the importance of the activity and telling them the importance of the topic. The students asked 2 minutes time to summarise the topic themselves before writing.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	1	1					1	1		

CO4: The students will be able to analyze the techniques required to create complex Embedded System.

References:

- Marilyn Wolf, “Computers as Components - Principles of Embedded Computing System Design”, Third Edition “Morgan Kaufmann Publisher, 2012
- <https://www.humber.ca/centreforteachingandlearning/instructional-strategies/teaching-methods/classroom-management-designing-the-environment/feedback-in-the-classroom/one-minute-papers.html>

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Degree, Semester& Branch: VII Semester B.E. ECE A

Course Code & Title: EC6703 Embedded and Real Time Systems

Name of the Faculty member: Mr.T.Ramprakash

Name of the Topic: Digital Camera

Name of the Innovative Practice: Animated Videos

Date & Time: 27.09.2019 & 07 Minutes

ICT Tool Used: LCD projector, Personal Computer, Speakers

Description:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

1. The students will be able to acquire better knowledge about the operation of Digital Camera.
2. The students will be able to understand the Bayer Pattern principle which is used in the Digital Camera.

Use of appropriate methods:

Justification for choosing Videos about Digital Camera:

Explaining the working principle of Digital camera and making the students to understand the concept of Bayer Pattern and explain they how millions of colours are stored and restored in a digital camera using chalk and talk is tedious process. So I chose video which clearly explains the following in an interesting manner

- How digital camera works?

- How digital still camera stores and restores millions of colours?
- What is the technique behind Bayer pattern and how it fits in digital still camera?

Effective presentation

Details of the Implementation:

The topic was first handled and many of the students were not able to imagine and visualise the working operation of digital still camera. Then the video was shown which made the students to understand well about the Bayer pattern and the working operation of the digital still camera.

Innovative Teaching Method Execution	
Digital Camera – Videos	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 80% of students answered correctly.

Reflective Critique:

- **Benefits:**
 - Some of the students were not able to imagine the working operation of digital still camera. But after watching the video, almost all the students were able to understand the concept of digital still camera.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	3	2	1					1	1		

CO 5: The Students will be able to apply the concepts of Embedded System Design in creating the model Real Time applications.

References:

- Marilyn Wolf, “Computers as Components - Principles of Embedded Computing System Design”, Third Edition “Morgan Kaufmann Publisher, 2012
- <https://www.khanacademy.org/science/electrical-engineering/reverse-engineering/digital-camera/v/what-is-inside-a-digital-camera-1-of-2-1>
- <https://www.youtube.com/watch?v=BYO1b-hvIuk>